Supplementary Notes for Granularity Ablation

1 Experimental Setup

We compare $N = 3$, 30, and 300 indivisible tokens per resource block using 100 sellers, each contributing 1–5 blocks, and 100–300 buyers. The initial resource spot price is 60, so the initial token spot price is $60/N$; actual payments include price impact and a 0.3% fee. Buyer budgets lie in $[60, 120]$. Physical demand is increased by 10% from the original distribution: $d_i = 11r_i/300$ blocks, where r_i is an integer from 20 to 100. The same supply, demand, budgets, and random procurement order are paired across token counts, with 30 seeds (42–71).

The displayed scenario uses a *translated constant-product AMM*, with pricing-depth factor $\beta = 3$. For actual cash X, actual token inventory Y, and their initial values X_0, Y_0, buying q tokens costs

$$c(q) = \frac{(X + \beta X_0)q}{(Y + \beta Y_0 - q)(1 - f)}, \qquad f = 0.003, \quad q \le Y.$$

Virtual offsets affect quotes only; they are not spendable cash or deliverable resources. All deliveries come from the original escrowed seller inventory. Positive offsets permit finite-price sellout of real inventory. This pricing rule differs from the original unshifted constant-product pool.

In one procurement session, each buyer acquires the largest affordable whole-token quantity up to $\lceil Nd_i \rceil$. The plotted resource-utilization outcome is *total purchased physical capacity divided by initial physical capacity*; every delivered fraction counts. The separate demand-matched outcome caps each buyer's acquired capacity by its actual demand.

2 Results and Analysis

Figure `ablation_allocation.pdf` reports the purchased-capacity outcome. At 300 buyers, the mean results are:

Tokens/block	Purchased capacity	Demand-matched capacity
3	95.82%	94.53%
30	100.00%	99.81%
300	100.00%	100.00%

Finer units reduce unused budget caused by an unaffordable next indivisible token and can reduce excess purchases beyond demand. In this scenario, no buyer is excluded because its initial cash cannot buy the first token. The results therefore support improved allocation under discrete purchase constraints, without establishing broader participation or a fairness gain.

The displayed $\beta = 3$ case comes from a sweep over $\{0, 1, 2, 3, 5\}$. A paired continuous-demand check gives 99.98% demand-matched allocation and 100% purchased capacity at 300 buyers. This separates exact demand-grid alignment from physical sellout; full allocation remains scenario-dependent.